Explicit upper bounds for the Erdős–Surányi function $g(n)$

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September 3, 2026

Abstract

For $n \geq 1$ let $g(n)$ be the least integer g such that for every set $\{a_1 < \cdots < a_n\}$ of integers greater than 1 and every $x \geq 0$, some g of the integers $x+1, \ldots, x+a_n$ have product divisible by $a_1 \cdots a_n$. Erdős and Surányi (1959) proved $g(3) = 4$ and $g(n) \geq (2 - o(1))n$, and Erdős asked (offering \$100) whether $g(n) \leq (2 + o(1))n$, or even $g(n) \leq 2n$ (Erdős Problem #708). To our knowledge no upper bound depending on n alone has been published. We prove

$$g(n) \;\leq\; n\big(\lceil \log_2 2n \rceil + \lceil \log_2 \lceil \log_2 2n \rceil \rceil + 2\big) \quad (n \geq 2), \qquad g(n) \;\leq\; \left\lceil 48\, n \sqrt{\tfrac{81 + \ln n}{\ln(81 + \ln n)}} \right\rceil,$$

and $g(n) \leq (2 + o(1))\, n \sqrt{\ln n / \ln \ln n}$. We also show that the conjectured bound holds for long intervals: if $a_n \geq 8n^3$ then $2n$ integers always suffice, so Erdős's question is reduced to the range $a_n < 8n^3$. All proofs are elementary and constructive: prime-power atoms of the a_i that are large relative to a_n are placed one per interval element by a per-prime greedy rule, the small atoms are packed into bins, and a counting argument bounds the number of bins. We also record $g(4) \geq 5$ and $g(5) \geq 6$ by exact computation.

1 Introduction

Gallai observed that among any a_2 consecutive integers one can always find two whose product is a multiple of $a_1 a_2$ (for $1 < a_1 < a_2$), but that there are $a_1 < a_2 < a_3$ and a run of a_3 consecutive integers among which no three have product divisible by $a_1 a_2 a_3$. Erdős and Surányi [1] studied the general situation. Following Erdős's formulation [2, §1], let $g(n)$ be the smallest integer such that, whenever $1 < a_1 < \cdots < a_n$ are integers and $x \geq 0$, there are integers $x < u_1 < \cdots < u_{g(n)} \leq x + a_n$ with $u_1 \cdots u_{g(n)} \equiv 0 \pmod{a_1 \cdots a_n}$. (Since enlarging B inside

*__Disclosure of the production pipeline.__ Every mathematical statement in this note was produced by automated systems inside an autonomous research pipeline: the proofs of Theorems 1.1 and 1.2 were found in two independent runs of OpenAI GPT-5.6 at "Pro" effort through the web interface (26 and 90 minutes of reasoning respectively, on 2026-09-03) and the proof of Theorem 1.3 by Qwen3.8-Max (Alibaba) through its web interface on the same day, all working from briefs written by an orchestrating agent (Claude Fable 5.1, Anthropic) that selected the problem, pinned the statement to Erdős's 1992 text, verified the lower-bound construction by exact computation, re-derived every step of both proofs, commissioned independent referee runs (Claude Opus 5 agents in fresh contexts, which also brute-forced the lemmas on hundreds of thousands of instances), and drafted this document. The three constructions were implemented and exercised on more than three million random instances, and every numerical inequality in the explicit parameter choices was checked by machine (scripts in the accompanying repository). No human supplied a definition, a lemma, a proof idea, or a correction. The problem itself remains open: the conjectured bound $g(n) \leq 2n$ is not proved here.

the interval preserves divisibility, "$g(n)$ integers" may be read as "at most $g(n)$ integers"; we use the latter, which also covers the case $a_n < g(n)$.) They proved $g(3) = 4$ and

$$g(n) \;\geq\; (2-\varepsilon)\, n \qquad (n > n_0(\varepsilon)), \tag{1}$$

the lower bound coming from the set of all products $p_i p_j$ of ℓ primes satisfying $2p_1^2 > p_\ell^2$, with the interval positioned so that a single integer u, divisible exactly once by each p_i, is the only common multiple of every $p_i p_j$ in it. Erdős [2] asked whether $g(n) < (2+\varepsilon)n$ for large n, "or perhaps even $g(n) \leq 2n$", and offered \$100 for a proof or disproof; the question is Problem #708 on the Erdős problems site [3]. Neither [2] nor [3] records an upper bound for $g(n)$, and we are not aware of one in the literature (the original paper [1], in Hungarian, was not accessible to us). Note that a bound in terms of n alone is not obvious: the trivial estimate $g \leq \sum_i \omega(a_i)$ (one interval element per prime factor of each a_i, see Lemma 2.3) depends on a_n, which is unbounded.

Throughout, $\mathrm{v}_p(k)$ is the exponent of the prime p in k, $\omega(k)$ the number of distinct prime factors of k, and $\Lambda(t) = \lceil \log_2 t \rceil$ is the least $r \geq 0$ with $t \leq 2^r$.

Theorem 1.1. *For every $n \geq 2$,*

$$g(n) \;\leq\; F(n) := n\big(\Lambda(2n) + \Lambda(\Lambda(2n)) + 2\big) \;=\; n \log_2 n + n \log_2 \log_2 n + O(n).$$

In particular $F(2) = 10$, $F(3) = 21$, $F(4) = 28$, $F(10) = 100$, $F(100) = 1300$, $F(1000) = 17000$.

Theorem 1.2. *For every $n \geq 1$, with $L = 81 + \ln n$,*

$$g(n) \;\leq\; \left\lceil 48\, n \sqrt{L/\ln L}\,\right\rceil, \qquad \textit{and} \qquad g(n) \;\leq\; (2+o(1))\, n \sqrt{\frac{\ln n}{\ln \ln n}} \quad (n \to \infty).$$

Theorem 1.1 gives the smaller value for every n below about 10^{104}; Theorem 1.2 is asymptotically stronger. Neither is close to the conjectured $2n$. For long intervals, however, the conjecture holds:

Theorem 1.3. *If $a_n \geq 8n^3$, then for every $x \geq 0$ some $2n$ of the integers $x+1, \dots, x+a_n$ have product divisible by $a_1 \cdots a_n$ (at most $2n$ are needed, and since $a_n \geq 2n$ the set may be padded); in fact Conjecture 6.4 below holds for such A. Consequently $g(n) \leq 8n^3$ as well, and the question whether $g(n) \leq 2n$ is equivalent to the same question restricted to $a_n < 8n^3$.*

The gap between $n\sqrt{\ln n/\ln\ln n}$ and $2n$ in the range $a_n < 8n^3$ is discussed in Section 6.

2 The common construction

Fix $A = \{a_1 < \cdots < a_n\} \subset \{2, 3, \dots\}$, put $m = a_n$, $I = \{x+1, \dots, x+m\}$ and $P = a_1 \cdots a_n$. We must find $B \subseteq I$ with $P \mid \prod_{b \in B} b$ and bound $|B|$. Note that B is a set: an interval element contributes its valuations once, however many a_i it may be a multiple of.

Lemma 2.1 (counting)**.** *For every integer $d \leq m$: $\#\{a \in A : d \mid a\} \;\leq\; \lfloor m/d \rfloor \;\leq\; \lfloor (x+m)/d \rfloor - \lfloor x/d \rfloor = \#\{b \in I : d \mid b\}$.*

Proof. The multiples of d in $\{1, \dots, m\}$ are $d, 2d, \dots, \lfloor m/d \rfloor d$, and the a_i are distinct elements of $\{2, \dots, m\}$. Any m consecutive integers contain $\lfloor (x+m)/d \rfloor - \lfloor x/d \rfloor \geq \lfloor m/d \rfloor$ multiples of d. □

Lemma 2.2 (whole interval)**.** *P divides $\prod_{b \in I} b$. Hence if $m \leq H$ then $B = I$ has $|B| = m \leq H$.*

Proof. $P \mid m!$ because the a_i are distinct elements of $\{2, \dots, m\}$, and $\prod_{b \in I} b = m!\binom{x+m}{m}$. □

Atoms, large and small. Fix an integer budget $H \geq 1$ and assume $m > H$; put $T = m/H > 1$. For $a \in A$ and $p \mid a$ call $q_p(a) = p^{\mathrm{v}_p(a)}$ an *atom* of a; the atoms of a are pairwise coprime and multiply to a, and a has $\omega(a)$ atoms. An atom is *large* if $q > T$ and *small* if $q \leq T$.

Lemma 2.3 (large atoms, one element each)**.** *Let $\mathcal{L}$ be the multiset of all large atoms of all $a \in A$ (atoms of different a are distinct members, so $r = |\mathcal{L}|$ counts with multiplicity). There is a set $U \subseteq I$ with $|U| \leq r$ such that for every prime p*

$$\mathrm{v}_p\Big(\prod_{u\in U} u\Big) \;\geq \sum_{q\in\mathcal{L},\ q \text{ a power of } p} \mathrm{v}_p(q).$$

More precisely, one can assign to each $q \in \mathcal{L}$ an element $\beta(q) \in I$ with $q \mid \beta(q)$ such that atoms of the same prime receive distinct elements; then $U = \beta(\mathcal{L})$.

Proof. Fix a prime p and list the large atoms of p as $p^{e_1}, \ldots, p^{e_s}$ with $e_1 \geq \cdots \geq e_s$. Each $a \in A$ has at most one atom of p, so the first j atoms come from j distinct elements of A, all divisible by p^{e_j}; by Lemma 2.1, I contains at least j multiples of p^{e_j}. Assign the atoms in this order, giving the j-th one a multiple of p^{e_j} in I different from the $j-1$ elements already used for p. Elements used for different primes may coincide; this only decreases $|U|$. For a fixed p the assigned elements are distinct, so their p-adic valuations add up to at least $\sum_j e_j$. □

Lemma 2.4 (assembly)**.** *Suppose the small atoms of all $a \in A$ (a multiset; atoms of different a are distinct members) have been partitioned into bins, each bin being a set of small atoms whose product f satisfies $f \leq T$ (a bin may contain atoms of different a). Let s be the total number of bins and $D = r + s$. If $D \leq H$, then there is $B \subseteq I$ with $|B| \leq D$ and $P \mid \prod_{b\in B} b$.*

Proof. Take U from Lemma 2.3, $|U| \leq r$. Order the bins $f_1, \ldots, f_s$ arbitrarily. Since $f_j \leq T = m/H$ and H is an integer, Lemma 2.1 gives at least $\lfloor m/f_j \rfloor \geq H$ multiples of f_j in I. When bin j is processed, the elements to avoid are those of U and the $j-1$ elements chosen for earlier bins, at most $r + s - 1 = D - 1 \leq H - 1$ in number; so an unused multiple v_j of f_j exists. Put $B = U \cup \{v_1, \ldots, v_s\}$. By construction $U \cap \{v_1, \ldots, v_s\} = \emptyset$ and the v_j are pairwise distinct, so $|B| \leq r + s = D$ and (this is the only place where the valuations of different elements are added)

$$\prod_{b\in B} b = \prod_{u\in U} u \cdot \prod_j v_j,$$

whose p-adic valuation is at least (sum over large atoms of p) + $\mathrm{v}_p(\prod_j f_j)$, which equals $\mathrm{v}_p(P)$ because every atom of every a is either large or lies in exactly one bin. □

Thus everything reduces to producing bins and a budget H with $D = r + s \leq H$. Theorems 1.1 and 1.2 do this in two different ways.

3 Proof of Theorem 1.1

Next-fit bins. For each $a \in A$ list its small atoms in increasing order of the prime and pack them by multiplicative next-fit: keep a current bin, initially empty (product 1); if multiplying the next atom keeps the product $\leq T$, add it; otherwise close the bin and start a new one containing that atom. Every small atom is $\leq T$, so every bin has product $\leq T$.

Lemma 3.1. *If $d_1, \dots, d_u$ are the bin products of one a in the order created, then $d_j d_{j+1} > T$ for $1 \le j < u$.*

Proof. Bin j was closed because the next atom q did not fit: $d_j q > T$. That atom is the first element of bin $j+1$, so $d_{j+1} \ge q$. $\square$

Lemma 3.2 (budget)**.** *Let $H, \kappa \ge 1$ be integers with*

$$\textit{(B1)}\ \ H \ge n(2\kappa+1), \qquad \textit{(B2)}\ \ H^{(\kappa+1)n} \le 2^{\kappa H}. \tag{2}$$

Then the next-fit construction with threshold $T = m/H$ produces $D = r + s \le H$ demands.

Proof. *Case $m^\kappa \le H^{\kappa+1}$.* Binning never increases the count, so $D \le W := \sum_{a\in A} \omega(a)$. Since every atom is at least 2, $2^{\omega(a)} \le a$, hence $2^W \le \prod_a a \le m^n$ and $2^{\kappa W} \le m^{\kappa n} \le H^{(\kappa+1)n} \le 2^{\kappa H}$ by (B2). So $W \le H$.

Case $m^\kappa > H^{\kappa+1}$. Then $T^{\kappa+1} = m^{\kappa+1}/H^{\kappa+1} > m$. Fix $a \in A$ with ℓ large atoms and u bins $d_1, \dots, d_u$. The large atoms and the products $d_1 d_2$, $d_3 d_4, \dots$ of consecutive bin pairs are pairwise coprime divisors of a, each exceeding T (Lemma 3.1). If $\ell + \lfloor u/2 \rfloor \ge \kappa+1$, the product of $\kappa+1$ of them would exceed $T^{\kappa+1} > m \ge a$ while dividing a; so $\ell + \lfloor u/2 \rfloor \le \kappa$, and since $u \le 2\lfloor u/2 \rfloor + 1$, $\ell + u \le 2(\ell + \lfloor u/2 \rfloor) + 1 \le 2\kappa+1$. Summing over a, $D \le n(2\kappa+1) \le H$ by (B1). $\square$

Lemma 3.3. *For $n \ge 2$ let $r = \Lambda(2n)$, $s = \Lambda(r)$, $C = r+s+2$, $H = nC = F(n)$ and $\kappa = \lfloor (C-1)/2 \rfloor$. Then* (B1) *and* (B2) *hold.*

Proof. (B1): $2\kappa + 1 \le C$. (B2), $n \ge 3$: then $2n \ge 6$, so $r \ge 3$ and $s \ge 2$; by definition $2n \le 2^r$, i.e. $n \le 2^{r-1}$, and $r \le 2^s$; also $s + 2 \le 2^s$ for $s \ge 2$. Hence $C = r+s+2 \le 2^{s+1}$ and $H = nC \le 2^{r-1} 2^{s+1} = 2^{C-2}$. Moreover $C \le 2(\kappa+1)$, i.e. $(C-2)(\kappa+1) \le \kappa C$, so $H^{(\kappa+1)n} \le 2^{(C-2)(\kappa+1)n} \le 2^{\kappa C n} = 2^{\kappa H}$. For $n = 2$: $r = 2$, $s = 1$, $C = 5$, $H = 10$, $\kappa = 2$, and $10^6 \le 2^{20}$. $\square$

Proof of Theorem 1.1. Let $H = F(n)$. If $m \le H$ use Lemma 2.2. Otherwise Lemmas 3.2 and 3.3 give $D \le H$, and Lemma 2.4 gives B with $|B| \le D \le F(n)$. $\square$

4 Proof of Theorem 1.2

Here the bins are formed by *maximal merging*: start with each small atom (of any a) in its own bin and, as long as two bins have product $\le T$, merge them. Each merge preserves the property "product $\le T$" and decreases the number of bins, so the process terminates, and at termination any two bins have product $> T$.

Lemma 4.1. *With $W = \sum_a \omega(a)$ and $D = r + s$ as before,*

$$\textit{(i)}\ \ D \le W, \qquad \textit{(ii)}\ \ D \ \le\ \frac{2n \ln m}{\ln(m/H)} + 1.$$

Proof. (i) is clear. (ii): any two bins have product $> T$, so at most one bin has product $\le \sqrt{T}$, and that bin still has product $\ge 2 > 1$. Hence, if $s \ge 1$, $P = \prod_{\text{large}} q \cdot \prod_j f_j > T^r\, T^{(s-1)/2}$ strictly, and $P \le m^n$; thus $r + (s-1)/2 < n \ln m / \ln T$ and $D = r + s < 2n \ln m / \ln T - r + 1 \le 2n \ln m / \ln(m/H) + 1$. If $s = 0$, $P > T^r$ gives $D = r < n \ln m / \ln T$. $\square$

Lemma 4.2. *For real $X > 1$, $\sum_{p\le X} 1/p \le e\ln(1+\ln X)$ (sum over primes).*

Proof. Put $\sigma = 1+1/\ln X$. For $p \le X$, $p^{\sigma-1} \le X^{1/\ln X} = e$, so $1/p \le e\,p^{-\sigma}$. Using $-\ln(1-t) \ge t$ and the finite Euler product, $\sum_{p\le X} p^{-\sigma} \le \ln\prod_{p\le X}(1-p^{-\sigma})^{-1} \le \ln\zeta(\sigma)$, and $\zeta(\sigma) \le 1 + \int_1^\infty t^{-\sigma}dt = 1+1/(\sigma-1) = 1+\ln X$. $\square$

Lemma 4.3 (average number of prime factors)**.** *For every real $t > 1$,*

$$\frac{1}{n}\sum_{i=1}^{n}\omega(a_i) \;\le\; \frac{\ln(m/n) + (t-1)\,e\ln(1+\ln m)}{\ln t}.$$

Proof. For every positive integer k, $t^{\omega(k)} = \prod_{p\mid k}\big(1+(t-1)\big) = \sum_{d\mid k,\ d \text{ squarefree}}(t-1)^{\omega(d)}$. All terms are nonnegative because $t > 1$. Summing over $k \le m$,

$$\sum_{k\le m} t^{\omega(k)} = \sum_{\substack{d\le m\\ d \text{ sqfree}}} (t-1)^{\omega(d)}\Big\lfloor\frac{m}{d}\Big\rfloor \le m\prod_{p\le m}\Big(1+\frac{t-1}{p}\Big) \le m\exp\Big((t-1)\sum_{p\le m}\frac{1}{p}\Big) \le m\,e^{(t-1)e\ln(1+\ln m)}$$

by Lemma 4.2. The a_i are distinct integers in $[2,m]$, so $\sum_i t^{\omega(a_i)} \le \sum_{k\le m} t^{\omega(k)}$, and by the AM–GM inequality $t^{\frac{1}{n}\sum_i\omega(a_i)} \le \frac{1}{n}\sum_i t^{\omega(a_i)}$. Taking logarithms gives the claim (note $m \ge n+1$, so $\ln(m/n) > 0$). $\square$

Lemma 4.4 (explicit parameters)**.** *Let $L = 81+\ln n$, $\ell = \ln L$, $R = \sqrt{L/\ell}$ and $H = \lceil 48nR\rceil$. If $m > H$ then $D < H$.*

Proof. Elementary facts: $\ell > 4$ (as $L \ge 81 > e^4$); $\ell \le L/4$ (the function $y/4 - \ln y$ is increasing and positive for $y \ge 81$); $\ln\ell \le \ell/2$ (as $y/2-\ln y > 0$ is increasing for $y \ge 4$); hence $R > 2$. Put $y = \ln m$, $z = \ln H$, $d = y - z > 0$, $c = z - \ln n$. Since $48nR > 1$, $H \le 96nR$, so $c \le \ln 96 + \ln R < 7 + \ell/2$ (using $\ln R = (\ell - \ln\ell)/2 \le \ell/2$) and $z < \ln n + 7 + \ell/2 < 2L$. Put $\Delta = 4\sqrt{L\ell} = 4R\ell \le 2L$ (as $4\ell \le L$).

Case $d \le \Delta$. Then $y \le 4L$. Apply Lemma 4.3 with $t = R$: $\ln R = (\ell-\ln\ell)/2 \ge \ell/4$ and $e\ln(1+y) \le e\ln(5L) < 3(\ell+3) < 6\ell$. Since $\ln(m/n) = d + c$,

$$\frac{W}{n} \le \frac{d+c+(R-1)\cdot 6\ell}{\ell/4} < \frac{4R\ell + 7 + \ell/2 + 6R\ell}{\ell/4} = 40R + \frac{28}{\ell} + 2 < 45R,$$

so by Lemma 4.1(i), $D \le W < 45nR < H$.

Case $d > \Delta$. By Lemma 4.1(ii), $D \le 2n\,\frac{z+d}{d} + 1 < 2n + \frac{2n\cdot 2L}{4\sqrt{L\ell}} + 1 = 2n + nR + 1 < 3nR < H$, using $R > 2$. $\square$

Proof of Theorem 1.2, explicit part. Let $H = \lceil 48nR\rceil$. If $m \le H$ use Lemma 2.2; otherwise Lemma 4.4 gives $D < H$ and Lemma 2.4 gives $|B| \le D < H$. $\square$

Proof of Theorem 1.2, asymptotic part. Fix $\varepsilon > 0$, put $Q_n = \sqrt{\ln n/\ln\ln n}$ and $H = \lceil(2+\varepsilon)nQ_n\rceil$; we show $D < H$ for all $m > H$ once n is large, which suffices as before. With $y = \ln m$, $z = \ln H$, $d = y - z$, $c = z - \ln n$ we have $z = \ln n + O(\ln\ln n)$, hence $\sqrt{z/\ln z} = (1+o(1))Q_n$ and $c = O(\ln z)$. Put $\Delta = \sqrt{z\ln z}$. If $d > \Delta$, Lemma 4.1(ii) gives $D/n \le 2 + 2z/d + 1/n \le (2+o(1))\sqrt{z/\ln z}$. If $d \le \Delta$, then $y = (1+o(1))z$; apply Lemma 4.3 with $t = \sqrt{z}/(\ln z)^2$, so $\ln t = (\frac{1}{2}+o(1))\ln z$: the three terms of the numerator divided by $\ln t$ are at most $(2+o(1))\sqrt{z/\ln z}$, $O(1)$ and $O(\sqrt{z}/(\ln z)^2)$ respectively, so $D/n \le W/n \le (2+o(1))\sqrt{z/\ln z}$. In both cases $D \le (2+\varepsilon/2)nQ_n < H$ for all $n \ge n_0(\varepsilon)$, so $g(n) \le \lceil(2+\varepsilon)nQ_n\rceil$ for $n \ge n_0(\varepsilon)$; as $\varepsilon > 0$ was arbitrary, $\limsup_n g(n)/(nQ_n) \le 2$. $\square$

Remark 4.5. No attempt was made to optimise the constant 48; the same argument with the case threshold and the prime-sum estimate tightened (e.g. an explicit Mertens bound in place of Lemma 4.2) gives a single-digit constant. Theorems 1.1 and 1.2 are proved by the same construction and differ only in how the number of bins is counted; Theorem 1.3 uses the same two ingredients (per-prime placement of large atoms, private multiples for small parts) with the budget $2n$ made possible by the length of the interval.

5 Long intervals: proof of Theorem 1.3

Assume $m = a_n \geq 8n^3$ and put $C = m/(2n)$. Then $C \geq m^{2/3}$ (equivalent to $m^{1/3} \geq 2n$), $C \geq 4n^2 \geq 2n$, and, since $m \geq 8n^3 \geq 4n^2$, also $C^2 = m^2/(4n^2) \geq m$. In the language of Section 2 we take the integer budget $H = 2n$ and threshold $T = C = m/H$ (note $m > H$).

Lemma 5.1 (two-bin packing)**.** *Let $C \geq 1$ be real and let $q_1, \ldots, q_r > 1$ be integers with $q_i \leq C$ and $P = q_1 \cdots q_r \leq C^{3/2}$. Then some sub-product s of the q_i (over a subset, possibly empty or everything) satisfies $P/C \leq s \leq C$; consequently the q_i can be partitioned into two classes with products $\leq C$.*

Proof. If $P \leq C$ take $s = P$. Otherwise $1 < P/C \leq \sqrt{C}$. If some $q_i > \sqrt{C}$, take $s = q_i$: then $s \leq C$ and $s > \sqrt{C} \geq P/C$. If all $q_i \leq \sqrt{C}$, multiply them in any order and stop at the first partial product $s \geq P/C$ (this happens at the latest at $s = P$); the previous partial product, possibly the empty product $1 < P/C$, was $< P/C$, so $s < (P/C)\sqrt{C} = P/\sqrt{C} \leq C$. In every case the complementary product is $P/s \leq C$. $\square$

Lemma 5.2 (split)**.** *Every $a \in A$ has at most one atom exceeding C, and admits a factorisation $a = d_1 d_2$ into coprime parts such that either $d_1, d_2 \leq C$, or d_1 is an atom $> C$ and $d_2 < 2n \leq C$.*

Proof. Two atoms $> C$ are coprime, so they would give $a > C^2 \geq m$, impossible. If $q = p^e > C$ is an atom, put $d_1 = q$, $d_2 = a/q < m/C = 2n$. Otherwise all atoms of a are $\leq C$ and their product $a \leq m \leq C^{3/2}$, so Lemma 5.1 splits them into two coprime classes with products $\leq C$. $\square$

Proof of Theorem 1.3. Split every $a \in A$ by Lemma 5.2; parts equal to 1 impose no condition and may be assigned to any element, so we discard them. This gives at most $2n$ *demands*, each either a *large* atom $> C$ or a *small* integer $\leq C$. (Lemma 2.4 with $H = 2n$ already gives $|B| \leq 2n$; we repeat the argument because Conjecture 6.4 asks for the explicit assignment.) Assign the large demands as in Lemma 2.3: for each prime p, the large atoms of p (one per a) receive distinct multiples of the required powers of p; different primes may share elements. Let U be the set of elements used, $|U| \leq$ number of large demands. Now assign the small demands one at a time: a small demand $d \leq C = m/(2n)$ has at least $\lfloor m/d \rfloor \geq 2n$ multiples in I by Lemma 2.1 ($2n$ is an integer), while the forbidden elements (U together with the elements already given to earlier small demands) number at most $2n-1$, because there are at most $2n$ demands in all and the current one is not yet placed; so d receives a multiple of its own, outside U and different from all other small representatives. Let B be the set of all elements used, $|B| \leq 2n$. At an element of U the demands present are prime powers of distinct primes, each dividing the element; at every other element of B exactly one demand d is present and d divides the element. Hence the product of the demands at any $b \in B$ divides b, and since the demands of each a multiply to a, $P \mid \prod_{b \in B} b$. This is exactly an assignment as in Conjecture 6.4. Finally, if $a_n < 8n^3$ then Lemma 2.2 gives a solution with at most $a_n < 8n^3$ elements, so $g(n) \leq 8n^3$. $\square$

6 Small values, obstructions and remarks

Proposition 6.1. *$g(4) \geq 5$ and $g(5) \geq 6$.*

Proof. Let $A = \{10, 12, 15, 20\}$ and $x = 50$, so $I = \{51, \ldots, 70\}$ and $P = 2^5 \cdot 3^2 \cdot 5^3$. No element of I is divisible by 25, so a solution contains at least three of the multiples $55, 60, 65, 70$ of 5; any set S of multiples of 5 in I has $\mathrm{v}_2(\prod S) \leq 3$ and $\mathrm{v}_3(\prod S) \leq 1$. Hence a further element with $\mathrm{v}_2 \geq 2$ and $\mathrm{v}_3 \geq 1$, i.e. a multiple of 12, is needed; the only multiple of 12 in I is 60, and using it still leaves $\mathrm{v}_2 \leq 3 < 5$. So at least five elements are needed, and $\{55, 63, 64, 65, 70\}$ works. For $A = \{5, 10, 12, 15, 20\}$ and $x = 50$ we have $P = 2^5 \cdot 3^2 \cdot 5^4$, so all four multiples of 5 are forced, supplying $\mathrm{v}_2 = 3$ and $\mathrm{v}_3 = 1$; two more elements are then needed for 2^2 and 3 (no element of I other than 60 is divisible by 12), and $\{55, 60, 63, 64, 65, 70\}$ works. Both minima were also confirmed by an exhaustive dynamic programme over capped valuation vectors, which further shows that no other x gives a larger minimum for these two sets. □

Remark 6.2 (why the obvious approaches fail)**.** (a) Choosing a multiple of each a_i and multiplying does not work, because the chosen multiples may coincide and a shared element contributes its valuations once; in the Erdős–Surányi family every a_i has the same unique multiple u. (b) One cannot in general match the $a_i \leq m/2$ to *distinct* multiples inside an interval of length m: van Doorn, Li and Tang [4] showed that an interval of length $2 \max A$ guarantees only $\min(n, \lceil 2\sqrt{n} \rceil)$ disjoint pairs (a, multiple of a), and this is optimal. (c) An induction that removes a_n, takes a solution B' for the rest and repairs it with two new elements fails: for $A' = \{2, 4, 8, 10, 12\}$, $I = \{1, \ldots, 16\}$, $B' = \{8, 12, 15, 16\}$ and $a_6 = 16$, the missing factor 2^4 cannot be supplied by two elements of $I \setminus B'$. Lemma 2.4 avoids all three pitfalls by giving every bin at least H candidates and letting different primes share an element only through the per-prime bookkeeping of Lemma 2.3.

Remark 6.3 (the gap to $2n$)**.** Both proofs pay one interval element per large atom and per bin. In the balancing region of Lemma 4.1 an a_i may have about $\sqrt{\ln n / \ln \ln n}$ atoms of "medium" size, none of which can be merged, and this is exactly the loss in Theorem 1.2. A linear bound would require medium atoms of *different* a_i to share representatives. A natural sufficient statement is the following.

Conjecture 6.4 (split assignment)**.** For every A and I as above one can choose for each $a \in A$ a factorisation $a = d_1(a)d_2(a)$ into coprime parts ($d_2 = 1$ allowed) and assign each of the at most $2n$ parts to an element $\beta(d) \in I$ so that for every $b \in I$ and every prime p, $\sum_{d:\beta(d)=b} \mathrm{v}_p(d) \leq \mathrm{v}_p(b)$.

The conjecture implies $g(n) \leq 2n$ (take $B = \beta(\text{parts})$), and Theorem 1.3 proves it whenever $a_n \geq 8n^3$. Since the capacity constraint couples two parts only if they share a prime, the assignment always exists when every $a \in A$ has at most two distinct prime factors (use the finest splitting and Lemma 2.3 prime by prime); a counterexample would need elements with three or more prime factors competing for composite multiples. An exact search found the assignment on the Erdős–Surányi instances with $\ell = 3, 4$, on the instances of Proposition 6.1, and on more than 10^6 random instances with $n \leq 6$ and $a_n \leq 210$; we found no counterexample.

References

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